MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE
(An Autonomous Institution)
(Sponsored by Matrusri Education Society, Affiliated to Osmania University)
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NBA Accredited B.E. Courses:- Civil, CSE, ECE, EEE, IT & Mechanical
Department of Computer Science & Engineering

Project Title: FairCred - Credit Engine using Explainable AI

Batch-ID: 19

Student Team Members:
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Guide: G Madhu , Assistant Professor, Dept. of CSE, MVSREC

Problem Statement:-

- Large population is underbanked/unbanked due to lack of formal credit history (CIBIL).
- Traditional credit models rely only on official financial data → excludes many users.
- Need an ethical, transparent, scalable model using alternate data to predict default probability.

Project Scope:-

- Use behavioral, digital, socio-economic, and contextual data.
- Build a synthetic dataset reflecting underbanked populations.
- Develop ML-based predictive credit scoring.
- Ensure privacy, explainability, and fairness.

Project Objectives:-

- Predict the probability of default (PD) without using traditional credit files.
- Integrate behavioral, digital, and socio-economic signals into a unified scoring pipeline.
- Build a privacy-preserving model with explainability through SHAP.
- Provide a practical framework usable by lenders through APIs and dashboards

Application Areas	Challenges
Digital Lending Platforms Used by NBFCs and fintech lenders for evaluating borrowers who lack formal credit history.	**Data Quality & Reliability** Alternative data signals may be noisy, sparse, or inconsistent across different user groups.
Underbanked / Informal Workforce Credit Helps extend credit access to gig workers, freelancers, small shop owners, and migrant laborers.	**Feature Interdependencies** Behavioral and contextual variables are correlated; maintaining realistic relationships requires complex conditional sampling.
Wallets and UPI Apps Can be integrated into apps like PhonePe-style wallets for personalized credit limits and loan eligibility.	**Bias & Fairness Issues** Behavioral features can unintentionally introduce socio-economic bias if not audited carefully.
Embedded Finance / API-based Credit Lenders can embed the risk scoring engine via APIs into partner apps (e-commerce, payments, telecom).	**Regulatory Compliance** Aligning with GDPR-like requirements and ensuring user data deletion, token integrity, and accountability is mandatory.
Policy Simulation & Economic Modeling Synthetic dataset allows regulators and researchers to simulate borrower behavior under different stress scenarios.	**Integration with Lender Systems** Legacy systems may not easily support API-based scoring or token verification.
Fraud Detection & Behavioral Verification Token-based verification prevents manipulation of user risk data and supports secure borrower identity profiling.	**User Trust & Adoption** Borrowers need to trust alternative-score systems, especially when data abstraction is unfamiliar to them.

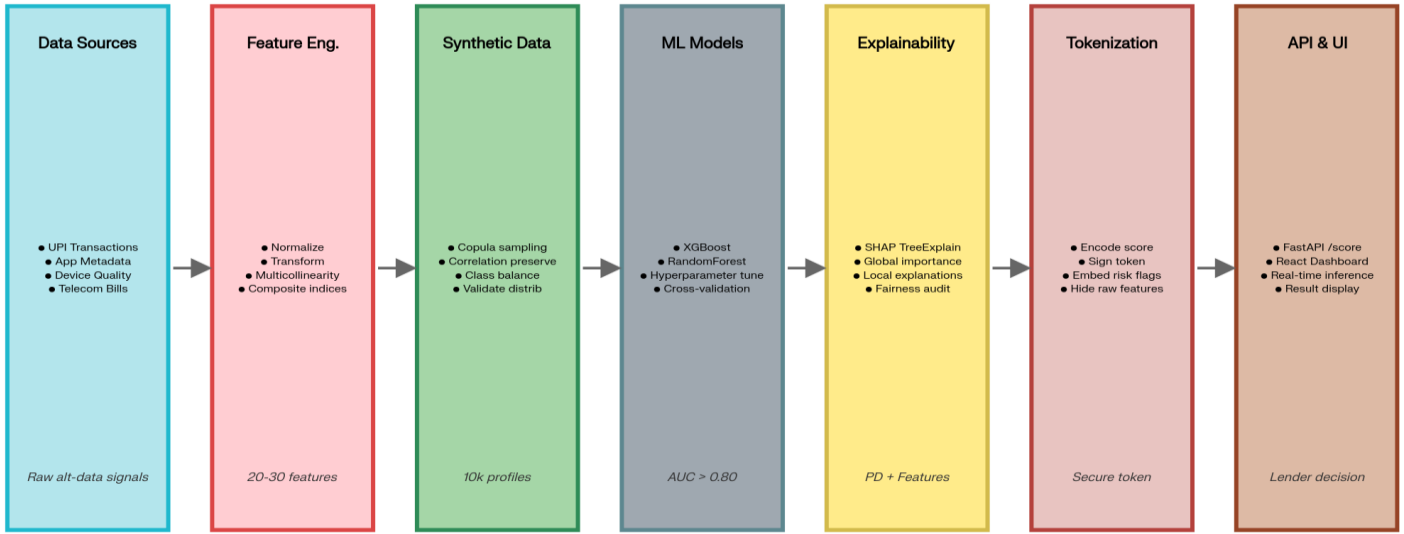
Literature Survey

SL	Major Reference(s)	Finding(s)
1	Research on alternative credit scoring using mobile and digital footprints (2019–2024)	Digital traces such as mobile recharge patterns, ride-hailing usage, and in-app payment history can significantly improve prediction of repayment behavior for new-to-credit users.
2	Feature engineering practices for credit risk and behavioral scoring	Grouping raw variables into interpretable indices (e.g., income tier, spending stability) makes models easier to explain and improves robustness across segments.
3	Use of tree-based ensembles (XGBoost, Random Forest) for credit risk modeling	Gradient boosting and random forests handle non-linear interactions between features and typically reach higher AUC than logistic regression on high-dimensional data.
4	Privacy-preserving synthetic data generation for financial datasets	Privacy-preserving synthetic data generation for financial datasets

Generic Architecture of the Existing Systems

FairCred System: Data Flow Architecture (7 Layers)

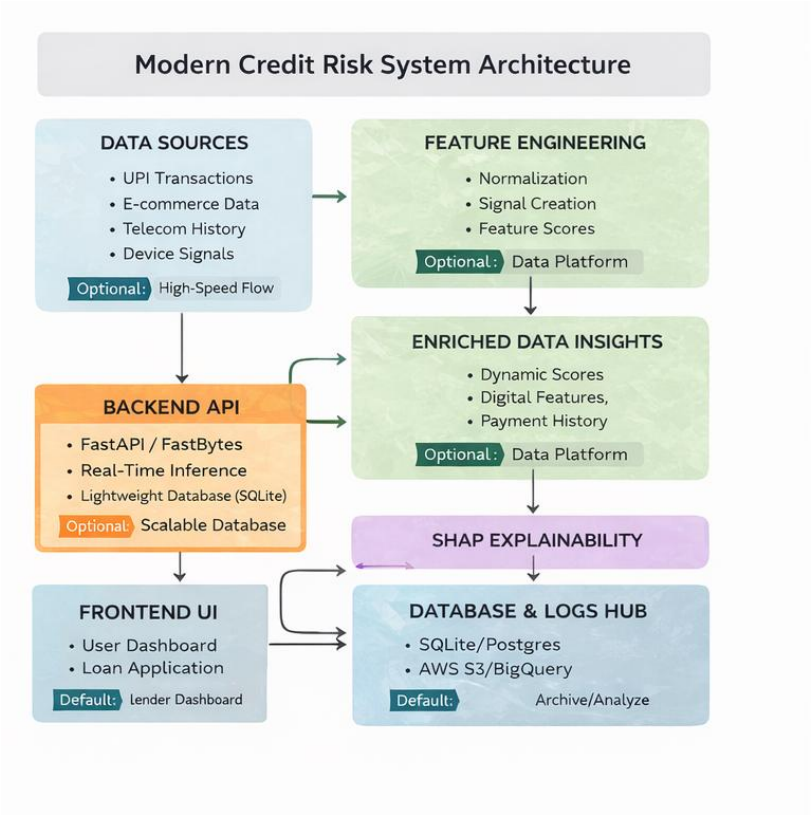
End-to-end credit risk modeling with explainability



Software / Hardware requirements for development / deployment

Tool / Technology	Usage
Frontend	ReactJS, HTML
Backend	Node.js
ML & Explainability	Python (XGBoost, Random Forest, SHAP), Pandas, NumPy,
Deployment	Docker

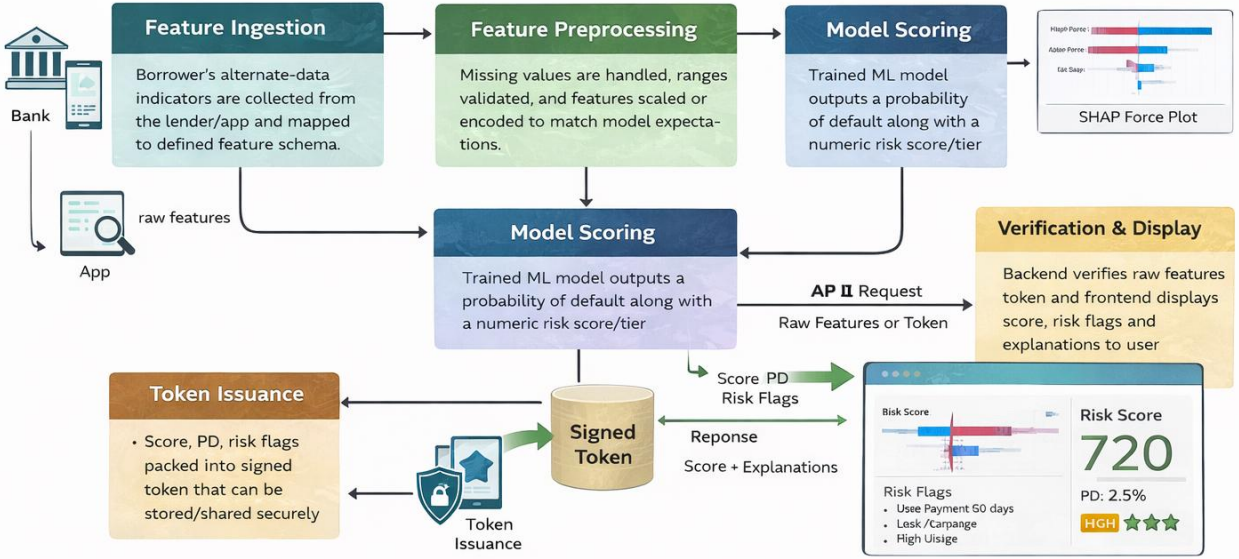
Proposed System Architecture



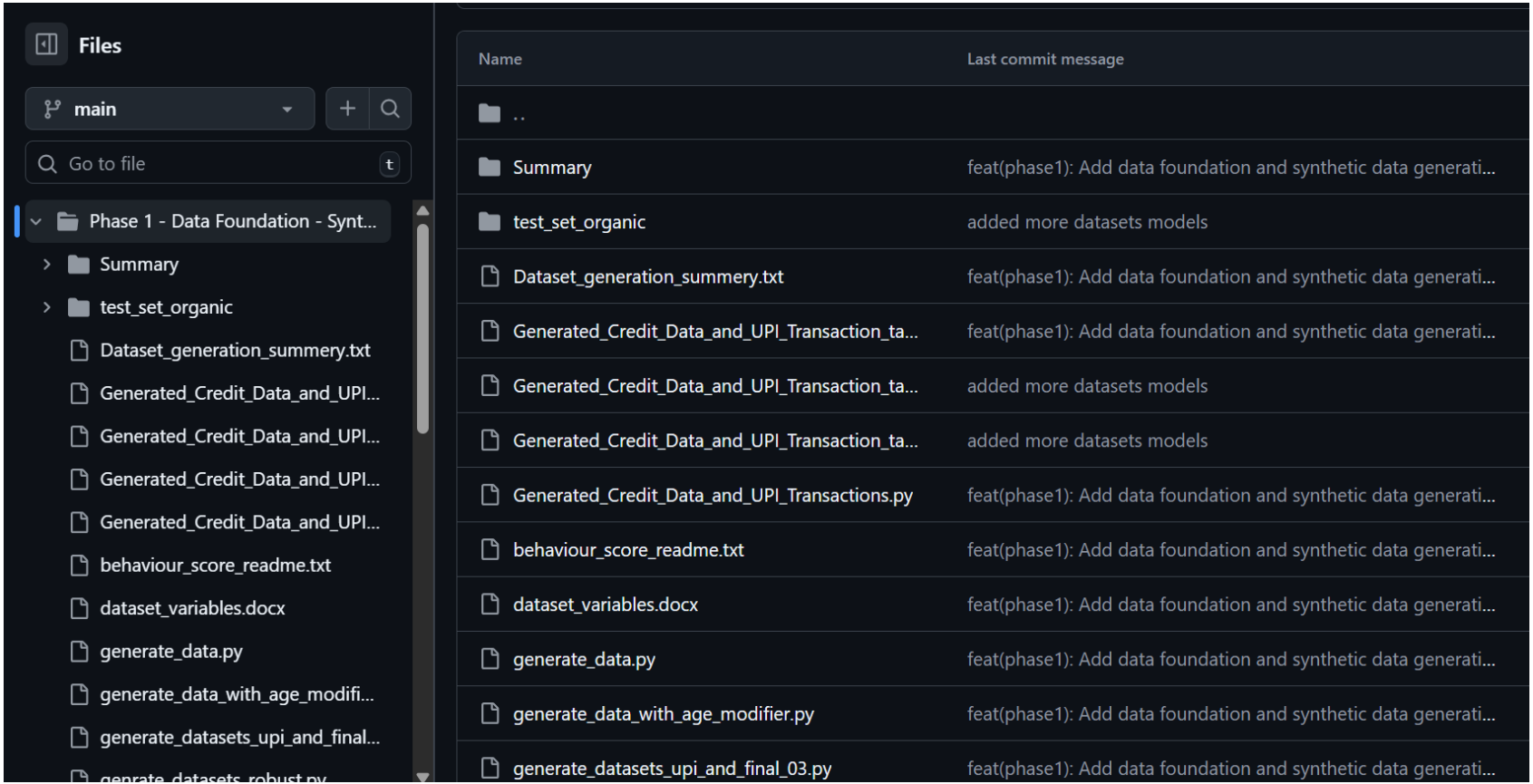
Module Description

Name of the module	Description of the module
Alternate Feature Design	Identify, define and categorize 20–30 alternate credit indicators from digital, behavioral and contextual sources; specify data types, valid ranges and transformations.
Synthetic Data Generator	Simulate realistic borrower profiles and default labels using parameterized statistical distributions and dependency rules; output clean tabular datasets for training and testing.
ML Modeling Engine	Implement preprocessing, train/validation/test splits and supervised models (Logistic Regression, RandomForest, XGBoost) to predict probability of default, and persist the best model.
Explainability Layer	Use SHAP to compute global feature importance and local explanations, generate visualizations and human-readable reason codes for each borrower's risk score.

Workflow of the system



Environmental Setup



Files

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- Phase 1 - Data Foundation - Synt...
- Phase 2 - Core Intelligence - ML ...
  - Model_01_making_logic.docx
  - confusion_matrix.png
  - predict_single_user_for_model_...
  - predict_single_user_for_model_...
  - shap_force_plot_high_risk.png
  - shap_force_plot_low_risk.png
  - shap_summary_plot.png
  - single_user_prediction_explanati...
  - train_and_explain_model_01.py
  - train_and_explain_model_02.py
  - train_and_explain_model_femal...

Name	Last commit message
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Model_01_making_logic.docx	feat(phase2): Add ML modeling, training, and explainability sc...
confusion_matrix.png	feat(phase2): Add ML modeling, training, and explainability sc...
predict_single_user_for_model_01.py	feat(phase2): Add ML modeling, training, and explainability sc...
predict_single_user_for_model_02.py	feat(phase2): Add ML modeling, training, and explainability sc...
shap_force_plot_high_risk.png	feat(phase2): Add ML modeling, training, and explainability sc...
shap_force_plot_low_risk.png	feat(phase2): Add ML modeling, training, and explainability sc...
shap_summary_plot.png	feat(phase2): Add ML modeling, training, and explainability sc...
single_user_prediction_explanation.png	feat(phase2): Add ML modeling, training, and explainability sc...
train_and_explain_model_01.py	feat(phase2): Add ML modeling, training, and explainability sc...
train_and_explain_model_02.py	feat(phase2): Add ML modeling, training, and explainability sc...
train_and_explain_model_female_tax_.py	feat(phase2): Add ML modeling, training, and explainability sc...
train_and_explain_model_with_clustering.py	feat(phase2): Add ML modeling, training, and explainability sc...

Document Structure

Chapter/Subchapter Sequence Number	Chapter/Subchapter Title
1	Introduction
2	Literature Survey
3	System Requirements & Design
3.1	Functional Requirements
3.2	Non-Functional Requirements
4	Data Modeling & Synthetic Data Generation
5	ML Modeling and Explainability

Thank You!

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